

DSA 8020 R Lab 7: Completely Randomized Designs

your name here, (the names of any collaborators)

Contents

The dataset `PlantGrowth` contains results from an experiment comparing yields (measured by the dried weight of plants) obtained under a control condition and two different treatment conditions.

Data Source: Dobson, A. J. (1983) *An Introduction to Statistical Modelling*. London: Chapman and Hall.

Let's load the data first:

Code:

```
data(PlantGrowth)
str(PlantGrowth)
```

```
## 'data.frame': 30 obs. of 2 variables:
## $ weight: num 4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.53 5.33 5.14 ...
## $ group : Factor w/ 3 levels "ctrl","trt1",...: 1 1 1 1 1 1 1 1 1 1 ...
```

1. Compute the treatment means and standard deviations

Code:

2. Make side-by-side boxplots by treatment

Code:

3. Write down the effects model and explain each term in the model, including the model assumptions regarding the random error.

Answer:

4. Perform an overall F -test using ANOVA. State the hypotheses, p -value, decision, and conclusion.

Code:

Answer:

5. Conduct pairwise comparisons using Tukey's HSD procedure. Interpret the results.

Code:

Answer:

6. Use a QQ plot to examine the normality assumption of the error.

Code:

Answer:

7. Make a residual plot to assess the equal variance assumption.

Code:

Answer:

8. Perform a Levene's test for equal variance.

Code:

Answer: